

**GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE, INDORE
END SEMESTER EXAMINATION**

B.E. — First Year (Mechanical Engineering)
Maximum Marks: 56 Time: 3 Hours

SET C

UNIT I — Mechanical Properties of Materials (14 Marks)

- Q1.** (a) Define elasticity. (2 Marks) (b) Define yield point. (2 Marks)
Q2. Explain the difference between elastic and plastic deformation. (3 Marks)
Q3. Describe hardness testing (Brinell, Rockwell, Vickers) with neat sketches. (7 Marks)

UNIT II — Theory of Machines (14 Marks)

- Q4.** (a) Define kinematic chain. (2 Marks) (b) Define inversion of mechanism. (2 Marks)
Q5. Explain the function of a cam and follower with classification. (3 Marks)
Q6. With neat diagram, explain the Scotch Yoke Mechanism. (7 Marks)

UNIT III — Fluid Kinematics (14 Marks)

- Q7.** (a) Define uniform flow. (2 Marks) (b) Define steady flow. (2 Marks)
Q8. Explain acceleration of fluid particles in Eulerian description. (3 Marks)
Q9. Derive the relationship between stream function and velocity components. (7 Marks)

UNIT IV — IC Engines & Automobiles (14 Marks)

- Q10.** (a) Define mean effective pressure. (2 Marks) (b) Define volumetric efficiency. (2 Marks)
Q11. Explain electronic fuel injection (EFI) system. (3 Marks)
Q12. Draw Diesel cycle on T-S diagram and explain each process. (7 Marks)